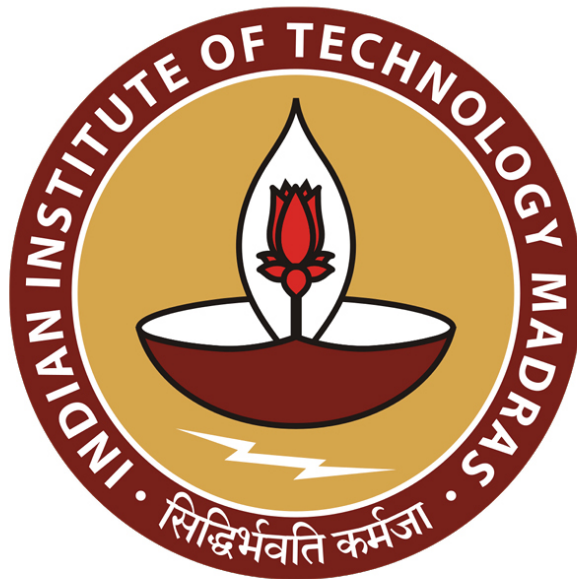
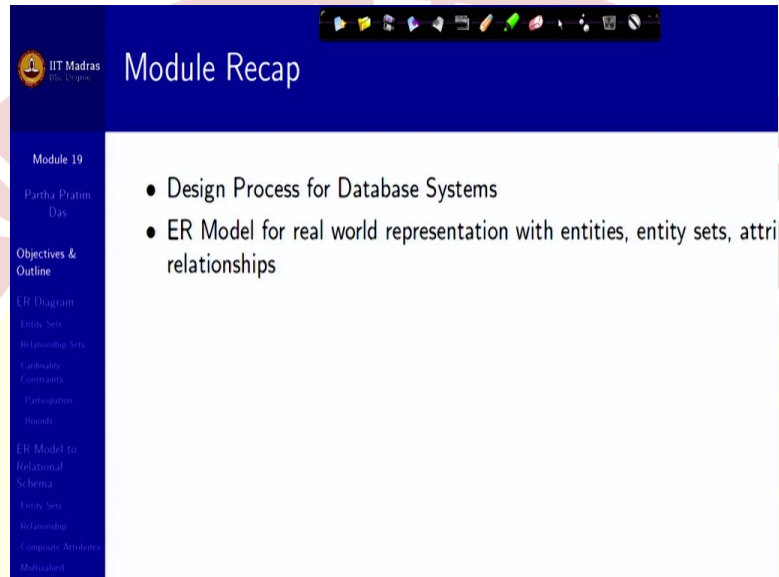




IIT Madras
ONLINE DEGREE

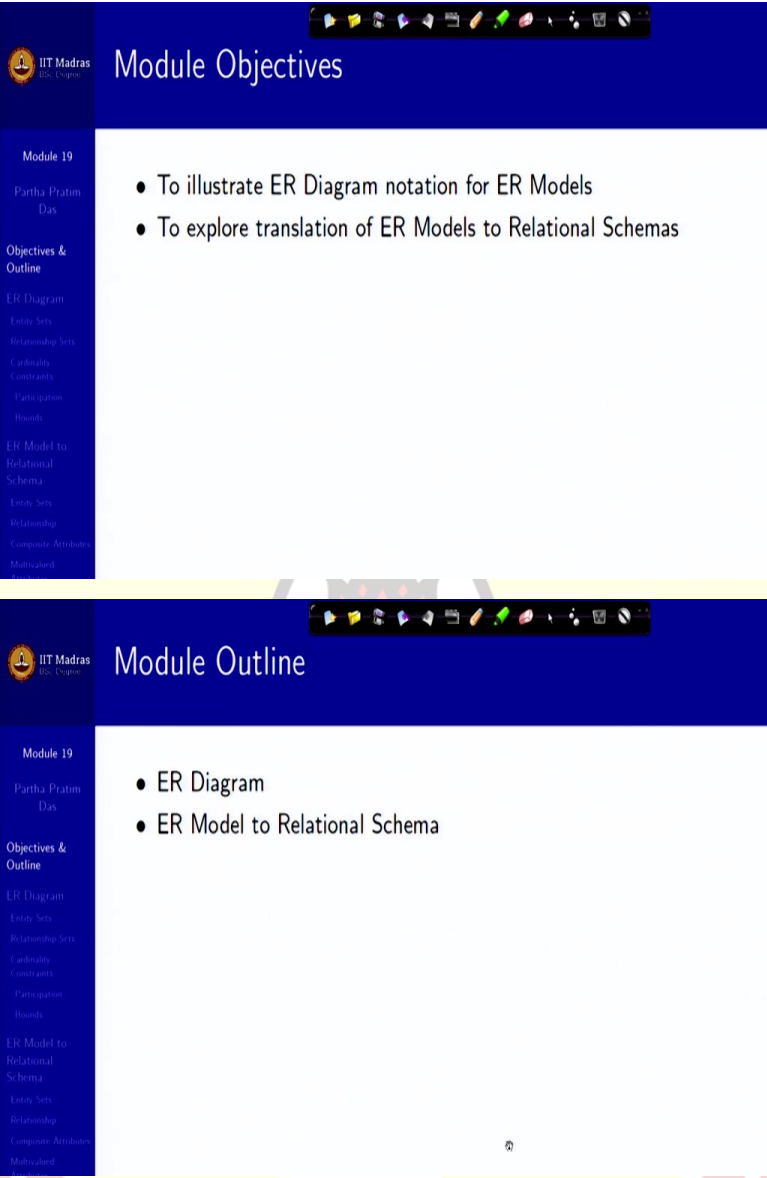
Database Management Systems
Professor Partha Pratim Das
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur
Entity- Relationship Model-2

(Refer Slide Time: 0:30)



Welcome to Module 19 of Database Management Systems course in the IIT Madras online B.Sc program. We have been discussing the database design process and we have taken a look at the basic premise of the design. And decided that we are initially focusing on the conceptual design or the logical model of the requirements. And for doing this, we are learning a framework or for this modeling which is entity relationship model in terms of entity sets attributes and relationships.

(Refer Slide Time: 1:00)



Module Objectives

- To illustrate ER Diagram notation for ER Models
- To explore translation of ER Models to Relational Schemas

Module Outline

- ER Diagram
- ER Model to Relational Schema

So, in the module in this current module we will learn the representation of these E-R models in terms of a diagram known as E-R diagram Entity Relationship diagram. And then we will see how this diagram can be translated into relational schemas or tables. So, these are the two main things to discuss in this.

(Refer Slide Time: 1:28)

ER Diagram

Entity Sets

- Entities can be represented graphically as follows:
 - Rectangles represent entity sets.
 - Attributes are listed inside entity rectangle.
 - Underline indicates primary key attributes.

instructor
<u>ID</u>
name
salary

student
<u>ID</u>
name
tot_cred

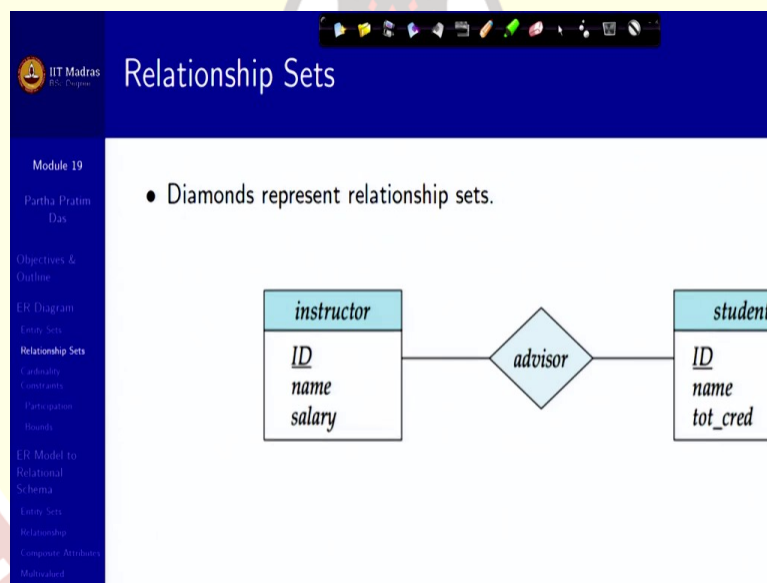
So, ER diagram ER diagram is a diagrammatic way to represent all the information that we have talked about so far. Now, this is something which evolved at the very early stages of database design and the notations were somewhat different at that time and at the end we will we have listed those alternate notations which you might find in several places. But over time it has evolved and with the introduction of a very powerful modeling framework known as UML unified modeling language.

These have got standardized now kind of in a certain diagrammatic way. So, for through this we will see how to represent attributes, how to represent entity sets, how to represent relationships,

how to represent cardinality and so on so forth a lot of things. So, entities are typically represented like this is a rectangle and which has the name of the entity set. And then you have the list of attributes one after the other.

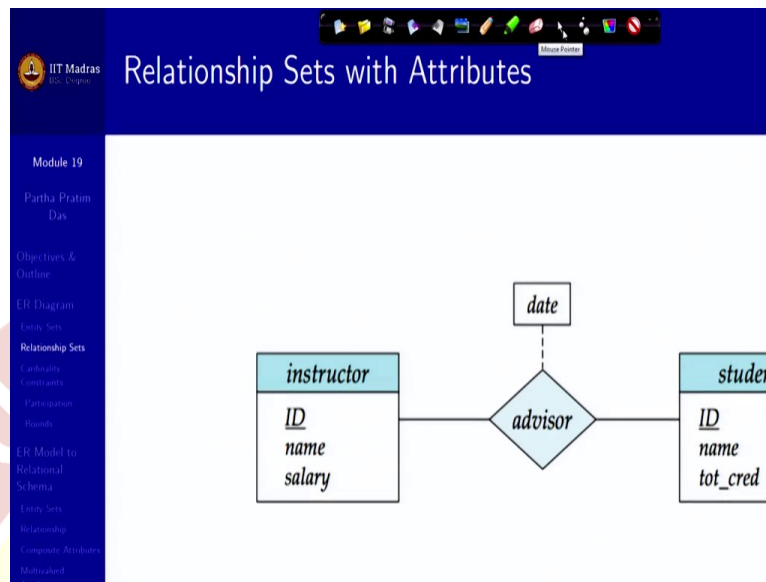
And the key of the entities set or the attributes that formed the primary key of the entity set are underlined. So, you can see what is the instructor and student and so on. And if you look up in terms of UML you will see that these are notations which are typically embedded in the class diagram set of diagrams in UML. And with the added information that every attribute there will not only have the name but will also have the type specified they go another level further. But we are we will for now just keep to the E-R model part of it. So, this is how you represent entity sets.

(Refer Slide Time: 3:28)



Now, between two entity sets you can have a relationship which are shown as diamond and in the middle of the diamond you give the name of the relationship and you draw connect the entity sets through lines. And implicitly it means that advisor is a relationship between instructor and student and the relationships unless otherwise specified will mean the relationship between the key of the entity set on left and the key of the entity set on right. So, advisor will have instructor.id and student.id this pair to work with.

(Refer Slide Time: 4:07)



It is possible that we talked about that it is possible that the relationship itself has an attribute. So, if it has if the advisor relationship has a date say from where the advising process has started. Then that attribute is directly connected to the relationship through a dashed line. Dashed line means that this is not a relationship line and because if you there is this is a diagrammatic language. So, the note the symbols and notations are very important.

For example, if you make this what happens if you make this solid line? If you make this a solid line then date will be that will mean that advisor has a relationship with the entity set date. Now, date will mean an entity set which has no attribute possible. I am saying there is entity set which has no attribute I have not yet decided on the attributes. But there is a relationship with date there is a concept called date through which it will have a relationship.

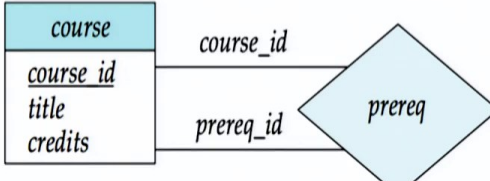
But that is not the case we use a dashed line the moment we use a dashed line it does not mean a relationship with an entity set it means that this relationship has an attribute. So, date now is interpreted as an attribute and it is an attribute of advisors. So, advisor will now become triplet instructor.id student.id date so remember.

(Refer Slide Time: 5:48)

 IIT Madras
Dr. C.V. Raman
Module 19
Partha Pratim Das
Objectives & Outline
ER Diagram
Entity Sets
Relationship Sets
Cardinality
Constraints
Participation
Bounds
ER Model to Relational Schema
Entity Sets
Relationships
Composite Attributes
Multivalued

Roles

- Entity sets of a relationship need not be distinct Each occurrence of a "role" in the relationship
- The labels "course_id" and "prereq_id" are called **roles**.



Be careful about the notation then their roles entity sets in the relationship may not be distinct. A entity set may have a relationship with itself for example course id and prerequisite we have seen every prerequisite is itself a course_id. So, prerequisite is a relationship between course and course where naturally. How do you distinguish them? You need to, so on the other side on one side you call it a course id on the other side you are calling it a prerequisite id.

You are giving a new name because it had two instances of the same relation. So, by name it is the same course.course_id, course.course_id. So, kind of here you make use of the renaming concept we had used. So, whatever is a prerequisite you call it a prerequisite_id. So, you say that an entity here in the relationship prereq is between a course_id which has a role of defining the course and a prereq_id which is actually a course_id, but has the role of specifying that what is the prerequisite. So, this is a typical kind of model.

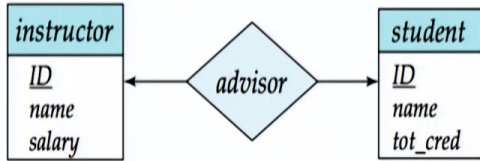
(Refer Slide Time: 7:08)

**IIT Madras**
B.Tech. Computer Science & Engineering

Cardinality Constraints

Module 19
Partha Pratim Das
Objectives & Outline
ER Diagram
Entity Sets
Relationship Sets
Cardinality Constraints
Participation
Bounds
ER Model to Relational Schema
Entity Sets
Relationship
Composite Attributes
Multivalued

- We express cardinality constraints by drawing either a directed line (—) "one," or an undirected line (—), signifying "many," between the relationship and the entity set.
- One-to-one relationship between an *instructor* and a *student* :
 - A student is associated with at most one *instructor* via the relationship *advisor*.
 - A *student* is associated with at most one department via *stud_dept*.



```
graph LR; instructor[instructor] <-->|advisor| student[student];
```

There could be cardinality constraints as we have seen. So, cardinality constraints are shown simply as if it is how the arrow ends. If it ends like this it is many if it ends like this it is one. If there is a arrow at the end then it is one, if it does not have an arrow at the end then it is many that is the basic meaning of the cardinality. So, you can very easily understand that it is saying that advisor is a relationship between instructor and student and it is one to one. Which means every instructor has only one student and every student has only one instructor that is what is being said.

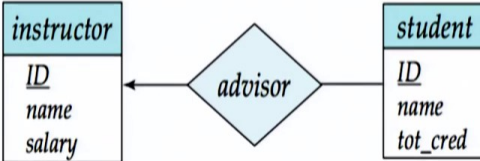
(Refer Slide Time: 7:50)

**IIT Madras**
B.Tech. Computer Science & Engineering

One-to-Many Relationship

Module 19
Partha Pratim Das
Objectives & Outline
ER Diagram
Entity Sets
Relationship Sets
Cardinality Constraints
Participation
Bounds
ER Model to Relational Schema
Entity Sets
Relationship
Composite Attributes
Multivalued

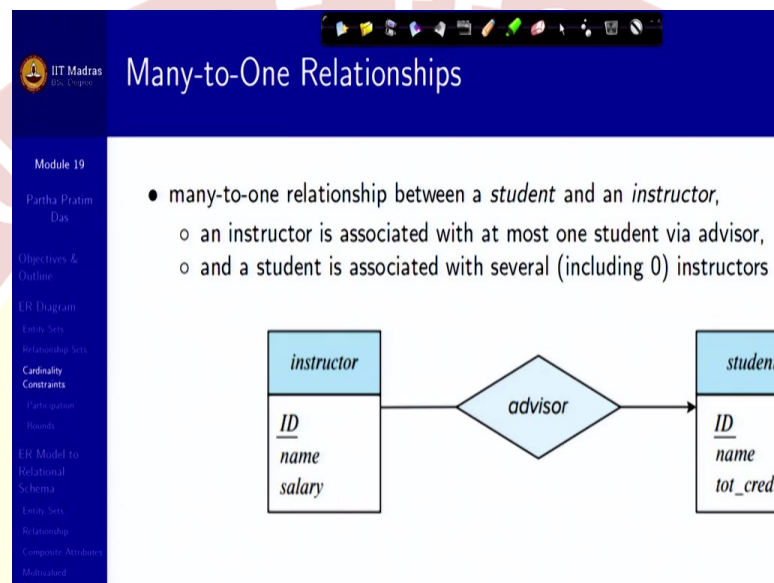
- one-to-many relationship between an *instructor* and a *student*
 - an instructor is associated with several (including 0) students via *advisor*
 - a student is associated with at most one instructor via *advisor*



```
graph LR; instructor[instructor] -->|advisor| student[student];
```

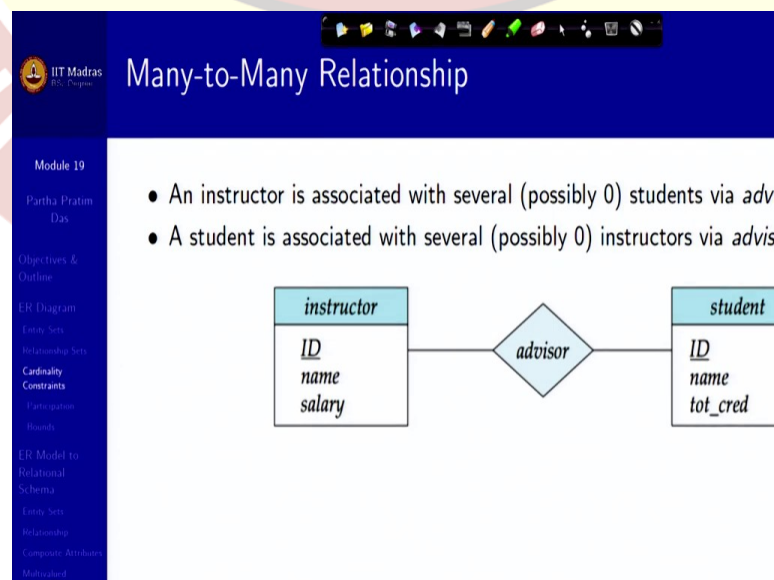

So, if I do which is which is more typical that it is one to many that every advisor every instructor will have many students as advising. Whereas every student will have a unique so you have to see which side that arrow is. So, arrow is on the left side. So, instructor is one student will be many.

(Refer Slide Time: 8:17)



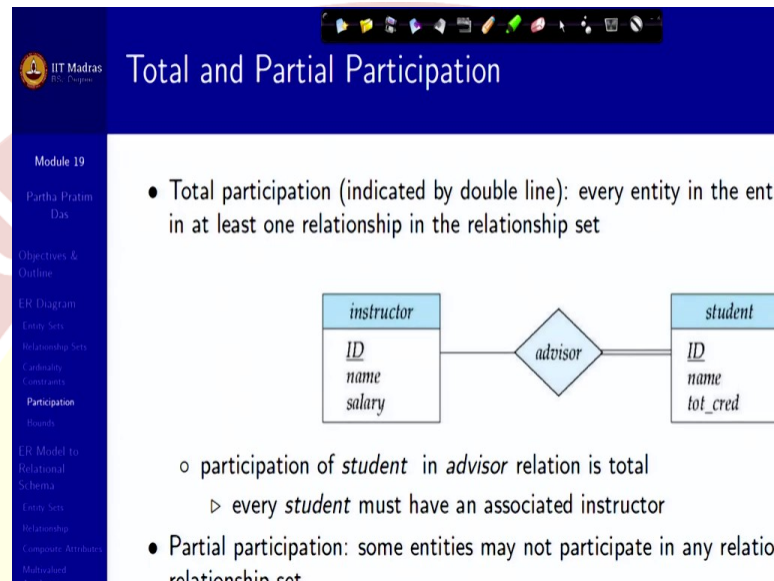
And obviously when I say one it is possible that it is not existence so it 0 is also it is 0 or 1 I should rather say this is many to one. Where this now says that a student may have multiple advisors but every advisor will advise only one student.

(Refer Slide Time: 8:39)



And you can have many to many where every advisor may have multiple students every student may have multiple advisors you can see that there is no arrow on either side. So, these are how the cardinalities are shown.

(Refer Slide Time: 8:49)



Then the question of total and partial participation we just talked about it in the last module. So, when you have a total participation of an entity set in a relationship you show that by a double line. So, the advisor line has a double line connection with the student. So, it means that every student must participate in the advisor relationship. In generalise what are you say that is it could be many to one means it could be 0 also.

But that is not possible in this case. Whereas or between the instructor and advisor there is only one single line which means that this participation is partial. So, an instructor may or may not feature in an advisor relationship. So, instructor may not have a student but a student must have a instructor. So, this whole idea is getting modelled through this representation of total and partial participation. We discussed total participation in context of weak entity set but it is a general concept which can be which will be represented and used in multiple ways.

(Refer Slide Time: 10:16)

Notation for Expressing More Complex Constraints

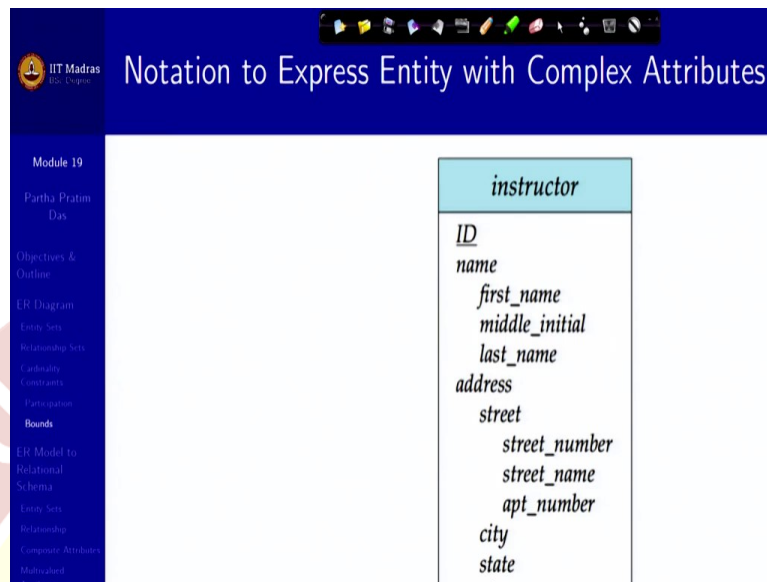
- A line may have an associated minimum and maximum cardinality, $l..h$, where l is the minimum and h the maximum cardinality
 - A minimum value of 1 indicates total participation.
 - A maximum value of 1 indicates that the entity participates in at relationship
 - A maximum value of * indicates no limit.

```
graph LR; instructor[instructor] ---|0..*| advisor{advisor}; advisor ---|1..1| student[student];
```

Now, it is possible that instead of just doing one to many or many to one and things like that you could actually specify the cardinality of a relationship on both sides of entity set. For example, here on advisor it is showing 1..1. So, this is a minimum to maximum. So, 1..1 it says that it has to be minimum 1 and it has to be maximum 1 which says that it is uniquely 1 which says that it is total that's it. Whereas on the instructor side it is saying it is 0..*, * means anything it is a wildcard no limit.

So, it says 0..* which means that an instructor may not have a student may have 1 student may have 10 students may have 1000 students everything is possible. So, this is a pure modeling of the meaning and it is this is therefore the moment you include 0 it means that it is possible that the instructor does not feature in the advisors. So, the participation is partial.

(Refer Slide Time: 11:26)



This is the notation for representing complex attributes. So, you write the composite attribute and then write the component attributes with a tab. So, name this reads that name is an attribute which comprise of first name, middle name, last name. Address is a composite attribute comprising of street, city, state, zip. State in turn is a composite attribute and so on. Then note another thing that in the phone number we have put the phone number within curly braces.

That means that, this is a multivalued attribute you remember single attribute composite attribute and single valued attribute multivalued attribute. So, here in this we are showing the notation for composite attribute as well as multivalued attribute. In the last if you see age you see a pair of parenthesis after it which is what which is a function notation what do we write like this function notation which means that this attribute has to be computed.

And we discussed that age is a derived attribute because it cannot be stored directly it will have to be computed from the date of birth. So, that information of derived attributes represented here. So, this is just one single slide which tells you what are the different ways information can be represented in the E-R diagram for attributes simple attributes, key composite attribute, multivalued attribute, derived attribute, everything is shown through this representation. So, just try to remember these.

(Refer Slide Time: 13:27)

**IIT Madras**
BSc. Program

Expressing Weak Entity Sets

Module 19
Partha Pratim Das
Objectives & Outline
ER Diagram
Entity Sets
Relationship Sets
Cardinality
Constraints
Participation
Bounds
ER Model to Relational Schema
Entity Sets
Relationships
Composite Attributes
Multivalued

- In ER diagrams, a weak entity set is depicted via a double rectangle
- We underline the discriminator of a weak entity set with a dashed line
- The relationship set connecting the weak entity set to the identifying is depicted by a double diamond
- Primary key for *section* – (*course_id*, *sec_id*, *semester*, *year*)



If you have to represent a weak entity set naturally what are the conditions you need a relationship between the strong entity set and weak entity set. So, first the representation of a weak entity set is done by using a rectangle for the entity set which has double line boundary. See course has single line boundary, section has double line boundary. So, that double line boundary says that a section is a weak entity set.

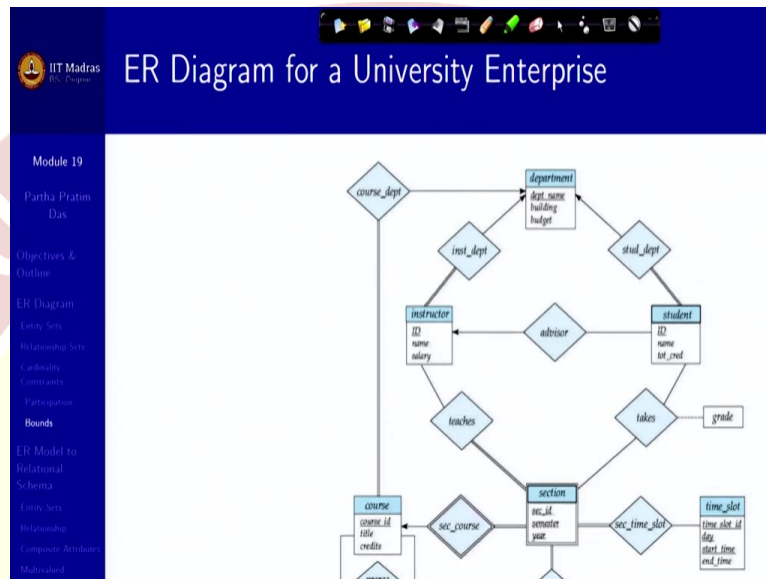
That immediately means that section must be associated with some strong entity set through a relationship. So, that relationship is *sec_course*. So, this relationship is linking a weak entity set to a strong entity set. So, to represent that the diamond of this particular relationship is also drawn with double lines. You see this diagrammatic language every way of drawing will mean something or the other.

So, we know that section is a weak entity set, course is a strong entity set, *sec_course* is a relationship from section to course to define the weak entity set properly. So, since and then you see that between *sec_course* and section the line connecting is double line. What does it mean? That this has a total participation. So, now you see that all of these conditions have been satisfied. So, what is remaining for the weak entity set I need to know the discriminator.

So, the discriminator is shown in the section entity set representation by doing hashed dashed underline of the attributes. So, section id semester and year this triplet is a discriminator. And this is underlined with dashed saying that this is a discriminator not a primary key. So, the

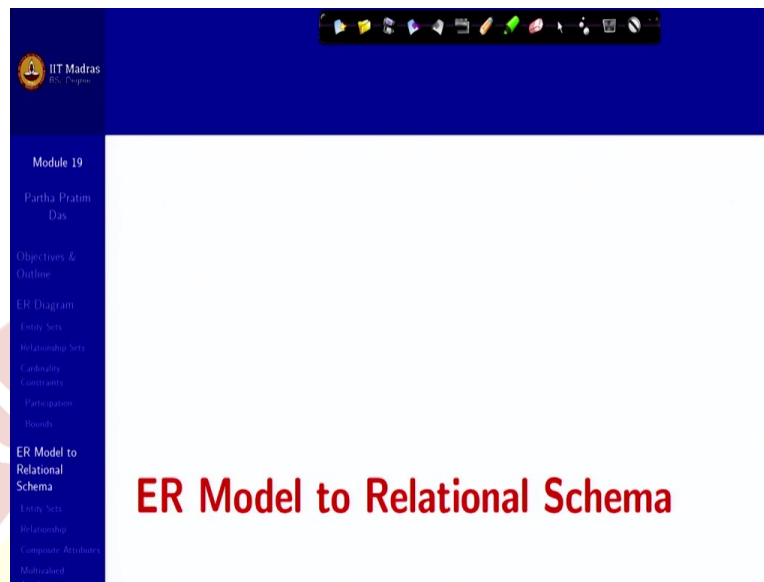
section course will associate sec_course will associate this with the strong entity set course whose primary key course id along with this discriminator that is course_id, section_id, semester and year, this four together will be the primary key of the weak entity set section.

(Refer Slide Time: 15:58)



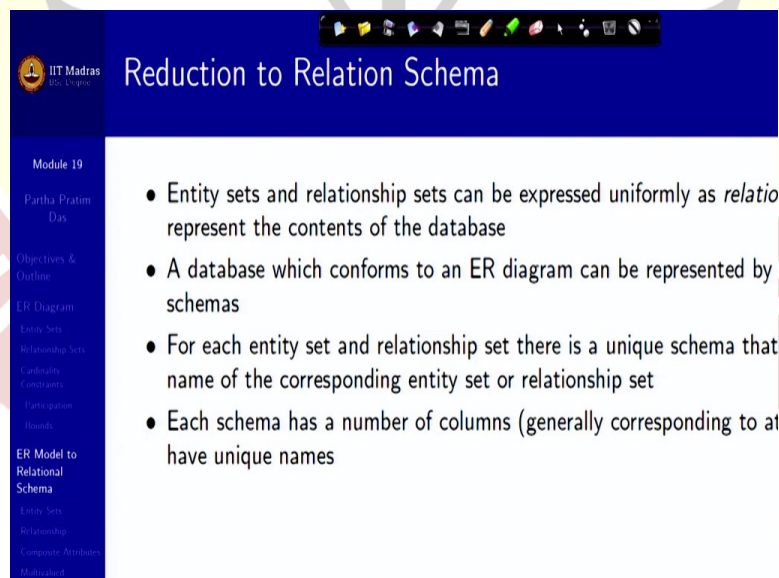
So, this is a more detailed diagram of the university representation that you had seen. So, earlier it was just showing well this is there that is there this kind of now it formally shows in terms of a ER model, you have to understand it very thoroughly and study it, get satisfied that all conditions that are being shown by the E-R model actually. What does it mean in the real world whether what whether it is kind of how meaningful it is and whether it is capturing the real world in the proper way or not.

(Refer Slide Time: 16:47)



So, we will move to the E-R model to relational schema that now we have the E-R model. So, we had a concept from that concept we form the attributes entity sets and relationships. Then we have represented that without ambiguity in terms of the E-R diagram. Now, we have to start creating the relational schema.

(Refer Slide Time: 17:15)



This reduction this is called reduction to relational schema. So, all that we need to do is primarily we need to know two things one is how to show the relational schema for entity set how to show the relational schema for a relationship.

(Refer Slide Time: 17:33)

 IIT Madras
Dr. Chandra

Representing Entity Sets

Module 19

Partha Pratim Das

Objectives & Outline

ER Diagram

Entity Sets

Relationship Sets

Cardinality

Constraints

Participation

Bounds

ER Model to Relational Schema

Entity Sets

Relationship

Composite Attributes

Multivalued

- A strong entity set reduces to a schema with the same attributes
student(ID, name, tot_cred)
- A weak entity set becomes a table that includes a column for the primary key of the identifying strong entity set
section(course_id, sec_id, sem, year)

```
graph LR; course[course] --- sec_course{sec_course}; sec_course --- section[section];
```

So, it is very simple for a relational schema for an entity set is you just take the name of the entity set as the name of the relation. And the list of the attributes as you have as a attributes. If it is a weak entity set then you have to do this process of going through the identifying relationship of the weak entity set the discriminator the primary key of the strong entity set. And get the primary key of the weak entity set put that there is list using the name of the weak entity set as the relationship. So, it is very obvious how these relationships are now coming out.

(Refer Slide Time: 18:14)

 IIT Madras
Dr. Chandra

Representing Relationship Sets

Module 19

Partha Pratim Das

Objectives & Outline

ER Diagram

Entity Sets

Relationship Sets

Cardinality

Constraints

Participation

Bounds

ER Model to Relational Schema

Entity Sets

Relationship

Composite Attributes

Multivalued

- A many-to-many relationship set is represented as a schema with attr primary keys of the two participating entity sets, and any descriptive relationship set.
- Example: schema for relationship set *advisor*
advisor = (s_id, i_id)

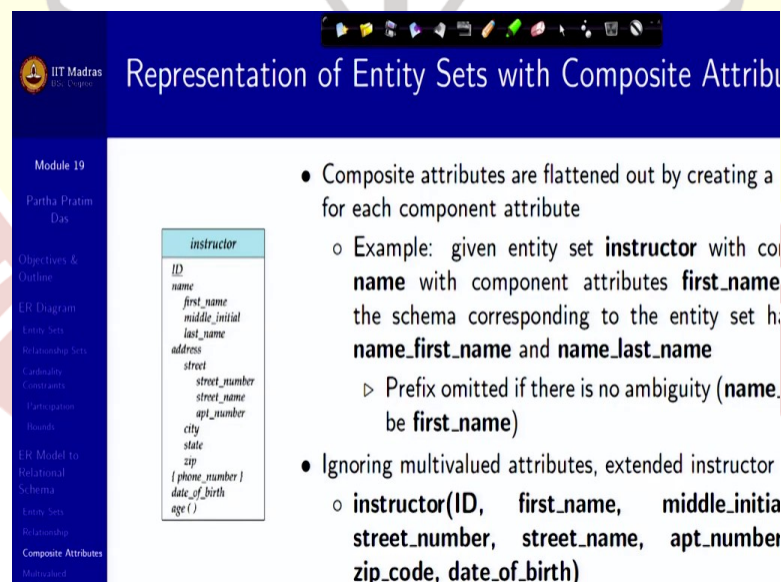
```
graph LR; instructor[instructor] --- advisor{advisor}; advisor --- student[student];
```

And you will see that things just go in a very smooth way. Then is a relationship in the relationship as I said by default the relationship between the primary keys of the relations that it is connecting it is associating. So, here in advisor you have the instructor id and the student id so you say it is s_id and i_id as names given you can write it as instructor_id, student_id also.

But the idea is you take the primary keys of two sides and put them as attributes. So, here you can see that basically it is a very simple thing you need to have what is a relation, a relationship is a name and a set of attributes and the information of the primary key. This is the fun three fundamental information for a relation the schema. So, the name is coming from the name of the entity set or the name of the relation and in terms of the attributes we are just figuring out how to get those attributes.

In case of strong entity set is just a list of attributes in case of weak entity sets is a list of attributes of the weak entity sets plus the primary key of the strong entity set through identifying relation. And in case of relationship, it is a set of primary keys put together of the two relationships that it is two entity sets that it is associating.

(Refer Slide Time: 19:43)



Representation of Entity Sets with Composite Attributes

- Composite attributes are flattened out by creating a name for each component attribute
 - Example: given entity set **instructor** with composite attribute **name** with component attributes **first_name**, **middle_initial**, and **last_name**, the schema corresponding to the entity set has attributes **name_first_name** and **name_last_name**
 - ▷ Prefix omitted if there is no ambiguity (**name_first_name** can be **first_name**)
- Ignoring multivalued attributes, extended instructor
 - **instructor**(ID, **first_name**, **middle_initial**, **street_number**, **street_name**, **apt_number**, **zip_code**, **date_of_birth**)

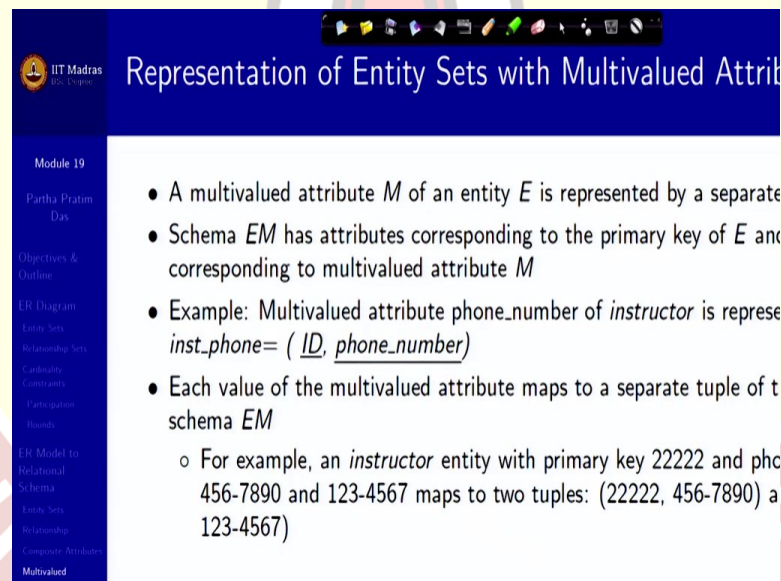
Now, how do you handle composite attributes? There could be two ways to handle that one is flattening. That you instead of getting into the

composite hierarchy as you have on left. So, instead of having name as an attribute will have first name, middle name, last name as three different attributes.

So, that could be the unifying schema the other approach could be that which in some cases will be used is in case of a composite attribute you will actually not represent the attribute here rather you will represent that attribute as a separate entity. You will say make an entity set for address then it becomes or say an entity set of name then it becomes simple attributes if you say name is an entity set then these are simple attributes.

And then you define a relationship between these two. So, it could be something of that but typically what we will do is flattening. So, if I have done that in the in the instructor design just ignoring the multivalued facet of the of the phone number you will see that we have the relational schema designed here.

(Refer Slide Time: 21:18)



Representation of Entity Sets with Multivalued Attributes

- A multivalued attribute M of an entity E is represented by a separate entity set.
- Schema EM has attributes corresponding to the primary key of E and corresponding to multivalued attribute M .
- Example: Multivalued attribute `phone_number` of `instructor` is represented by entity set `inst_phone = (ID, phone_number)`.
- Each value of the multivalued attribute maps to a separate tuple of the schema EM .
 - For example, an `instructor` entity with primary key 22222 and phone numbers 456-7890 and 123-4567 maps to two tuples: (22222, 456-7890) and (22222, 123-4567).

Now, how to handle multivalued now if you have to handle multiple values naturally you cannot put it right there. So, what you will have to do is you will have to create a separate entity set and have a kind of relationship with the original entity set. So, that we can say that in the multivalued phone number what do we know it is between the phone number and the instructor. What is the type of relationship it is one to many.

That is one instructor can have multiple phone numbers but one phone number cannot belong to multiple people. So, if I take the phone number phone number by itself is a key. So, I can

represent it in a table in a relationship I am sorry in an entity set with the id of the person who holds that phone number that becomes a relationship of using number which number am I using.

So, that could be a way to so that is why you see that `inst_phone` is a doublet of id and phone number by which we can now have multiple phone numbers associated with this. But obviously this means that they will I mean while we want to extract phone number, they will have to be some additional logic in the processing. Because it is no more just a flat attribute but it is coming through a relationship in the process.

(Refer Slide Time: 23:18)

Redundancy of Schema

- Many-to-one and one-to-many relationship sets that are total on the "many" side, contain an extra attribute to the "one" side
- Example: Instead of creating a schema for relationship set *inst_dept*, *dept_name* to the schema arising from entity set *instructor*

The diagram shows three entity sets: *instructor*, *department*, and *student*. The *department* entity has attributes *dept_name* (underlined), *building*, and *budget*. There are two relationship sets: *inst_dept* (many-to-one) and *stud_dept* (one-to-many). Handwritten notes include "(dept_name, I.D.)" and "(dept)" with arrows pointing to the *dept_name* attribute and the *stud_dept* relationship respectively.

Now, many to one and one to many relationship sets that are total on the many side can be represented by adding an additional attribute. For example, what we are saying say we have here let us understand this diagram again. Three entity sets instructor, department, and student. Three entity sets instructor, department, student. And three relationships one is advisor which is many to one from student to instructor.

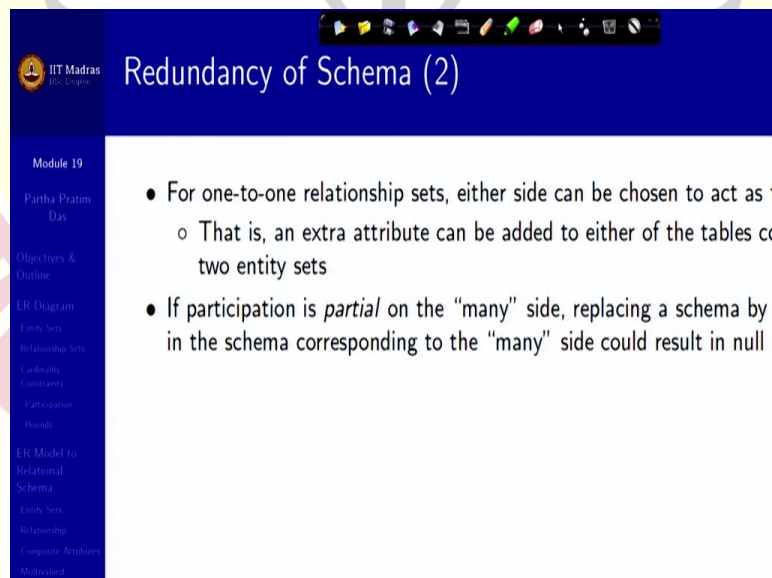
The other is the student department that is which department does a student belong to this is also many to one. But with the specificity that on the student side this is total which means that I cannot have a student who does not have a department is a total part. So, every student must be there in the *stud_dept* relationship which means that no student can be there without a department.

Then inst_dept is a similar relationship but between the instructor and the department. Again, this is also a total thing. So, if every instructor is so what is say what is the stud_dept relationship this relationship is if I when I put it into the relational schema it becomes department name and S_ID where both of them together is a key that is this relationship. And this one is department name and I_ID both of them are key.

Now, the point is since from the many side this is total which means that for every I_ID there is an entry in this and it is many to one. So, that entry is unique. Two things one is every I_ID will be there in this and it will be there only once. Because it is many to one it can have only one department if that be the case then the question is do we need this relationship or simply I can take the department name and put it as an attribute here.

Similar argument will say do we do I need this or I can take it and put it just like here. So, I can just extend the instructor and student entity sets or relationships by an additional attribute department name and get rid of these two relationships. Which will be two different tables like this altogether reduces my overhead of doing traversal join of relationships in answering the query.

(Refer Slide Time: 27:08)



IIT Madras

Redundancy of Schema (2)

Module 19
Partha Pratim Das

- For one-to-one relationship sets, either side can be chosen to act as t
 - That is, an extra attribute can be added to either of the tables corresponding to the two entity sets
- If participation is *partial* on the "many" side, replacing a schema by a schema in the schema corresponding to the "many" side could result in null values

So, that is a some of the considerations to scrutinize after a basic design has been made.

(Refer Slide Time: 27:19)

**IIT Madras**
BSc. Computer Science

Redundancy of Schema (3)

Module 19
Partha Pratim Das
Objectives & Outline
ER Diagram
Entity Sets
Relationship Sets
Cardinality Constraints
Participation Constraints
ER Model to Relational Schema
Entity Sets
Relationships
Composite Attributes
Multivalued

- The schema corresponding to a relationship set linking a weak entity identifying strong entity set is redundant.
- Example: The *section* schema already contains the attributes that *sec_course* schema

```
graph LR
    course[course] --- sec_course{sec_course}
    sec_course --- section[section]
```

So, this is also something that you can look at you can take this also is from many to one. This is on the section side this is total. So, section will be there in *sec_course* only once it will be unique and every section will be there. So, we can include *course_id* in the section that is in terms of modeling the weak entity set. Now, we are saying that weak entity sets discriminator then the identifying relation then the key primary key of the strong entity set. After all these I can just take the key of the strong entity set and put it in the weak entity set to make it a strong one and get rid of the identifying relationship.

(Refer Slide Time: 28:16)

**IIT Madras**
BSc. Computer Science

Module Summary

Module 19
Partha Pratim Das
Objectives & Outline
ER Diagram
Entity Sets
Relationship Sets
Cardinality Constraints
Participation Constraints
ER Model to Relational Schema
Entity Sets
Relationships
Composite Attributes
Multivalued

- Illustrated ER Diagram notation for ER Models
- Discussed translation of ER Models to Relational Schemas

Slides used in this presentation are borrowed from <http://db-book> permission of the authors.
Edited and new slides are marked with "DDN"

So, if you now go back and see the schema that exist in that we had been using you will see that because that was a relational schema. So, now we are slowly evolving and trying to come to that schema. So, you will see that this kind of considerations have gone in terms of first designing it, refining it and then removing the redundancy in a full-blown form. So, this brings us to the end of discussions in this module. We have discussed the basic E-R diagram and translation of E-R models to relational schema and we will continue in the next module. Thank you very much for your attention and see you in the next module.

